

Calibrated Perplexity for Robust AI-Generated Text Detection Across Different Domains

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Abstract. Many machine-generated text detectors use perplexity for prediction, however, their performance degrades substantially under domain shift, where a model trained on one domain (e.g., news) fails on another (e.g., scientific articles). In this paper, we propose an ensemble-based approach for robust AI-generated text detection that does not require knowledge of the test text’s domain. Our method trains separate statistical n-gram models on human-written texts from five distinct domains. For inference, we combine their predictions using three different strategies: (1) simple majority voting, (2) native domain detection based on minimal perplexity, and (3) weighted voting by TPR–FPR. Experiments on a large-scale Russian-language dataset demonstrate that all ensemble strategies achieve balanced performance (accuracy=0.76, F1=0.75), substantially outperforming single-domain models applied out-of-domain. Simple majority voting reaches 0.79 human recall and 0.72 AI recall, while native domain detection achieves the highest human recall (0.82). Our research confirms that domain shift is a critical challenge for perplexity-based detection, and that ensemble methods offer a practical, interpretable, and domain-robust solution.

Keywords: AI-generated text detection · Perplexity calibration · Cross-domain detection · Ensemble methods · Genre adaptation.

1 Introduction

The widespread accessibility of Large Language Models (LLMs) has transformed the landscape of text generation, offering unprecedented capabilities while simultaneously creating significant challenges. Machine-generated text now permeates academic writing, journalism, social media, and other domains, raising concerns about academic dishonesty, misinformation, and erosion of trust in digital content [15], [7]. These concerns have motivated extensive research into robust methods for detecting AI-generated content.

Current detection approaches fall into several categories: fine-tuned classifiers, statistical feature extraction, and zero-shot methods. Perplexity is a statistical measure of how “surprising” a text is to a language model, and it has

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proven to be particularly useful and computationally efficient [3]. The underlying assumption is straightforward: LLMs trained to maximize likelihood generate text with lower perplexity compared to human writing, as human authors exhibit greater variability and unpredictability.

However, perplexity is not an absolute measure. Its value depends heavily on the domain (e.g., news, scientific articles, social media) and language of the text. A language model fine-tuned on one domain often performs poorly on another, rendering detection unreliable in real-world scenarios where the target domain is unknown. Recent work has highlighted this domain shift problem [6] and proposed various solutions, including zero-shot detectors such as binoculars [7], which use scores ratios from two different LLMs for calibration, and cross-domain adaptation techniques [5].

Despite these advances, the problem of calibrating perplexity across domains remains insufficiently addressed. Existing methods either require knowledge of the test domain or rely on large neural models that are computationally expensive and not always interpretable. We argue that a lightweight, interpretable approach based on domain-specific calibration can offer comparable robustness while being more accessible and easier to deploy.

In this paper, we propose a novel method, called *calibrated perplexity*, which normalizes perplexity scores against domain-specific baselines. Our approach consists of three stages: (1) training separate KenLM n-gram models on human-written texts from different domains, (2) calibrating classification thresholds for each domain, and (3) combining their predictions using ensemble strategies that require no prior knowledge of the test text’s domain. The core novelty lies in transforming a domain-dependent score into a stable feature for classification through domain-specific calibration and ensemble aggregation.

Our contributions are threefold:

1. We demonstrate that optimal perplexity thresholds vary substantially across domains, confirming the need for calibration.
2. We propose three ensemble strategies for combining domain-specific models without requiring domain labels at test time.
3. We provide empirical evidence that ensemble methods outperform single-domain models on out-of-domain data, achieving robust detection across diverse domains.

2 Related Work

2.1 Perplexity-Based Detection

The problem of distinguishing human-written from machine-generated text has been extensively studied in recent years. Early approaches relied on supervised classifiers trained on statistical features such as n-gram frequencies, syntactic patterns, and stylometric indicators. Gehrmann et al. [3] introduced GLTR, a visual analysis tool that ranks token probabilities from a language model and

showed that perplexity and token probability distributions are effective statistical signals for detection.

Perplexity has since become one of the most commonly used signal for zero-shot detection. However, perplexity values vary substantially from model to model, so raw scores cannot be compared directly and need to be calibrated [10]. One of the most popular perplexity-based method is Binoculars, which uses a ratio of perplexity scores from two different language models (a base model and a larger reference model) to calibrate predictions [7]. This method achieves strong zero-shot performance and is particularly effective at detecting text from previously unseen generative models. Binoculars is conceptually related to our work, as both methods aim to normalize perplexity scores to make them comparable across different texts and domains. However, our approach uses domain-specific n-gram models as baselines rather than pairs of neural language models. This choice offers a more interpretable and computationally efficient solution.

2.2 Cross-Domain and Multilingual Detection

While such methods calibrate perplexity across models, they do not explicitly account for variation across text domains. Domain shift has long been identified as a key obstacle to robust detection in real-world applications: Zellers et al. [16] showed that detectors trained on one domain often perform poorly when applied to others, and He et al. [8] introduced the M4 benchmark for systematically evaluating detectors across multiple generators, domains, and languages. Gritsai et al. [6] explicitly demonstrated that models fine-tuned on one domain suffer significant performance degradation when applied to others, and further showed that n-gram-based statistical methods, including perplexity computed from n-gram models, achieve competitive results for detecting artificially generated fragments in academic documents [5]. These findings motivate our choice of n-gram models as a lightweight and domain-adaptable detection framework.

For Russian specifically, the RuATD Shared Task 2022 [11] provided the first benchmark for Russian, featuring texts from 13 generative models across multiple domains, and showed that while detection is possible, models often struggle with generalization. Building on RuATD, the authors introduced the CoAT corpus [12], a larger and more diverse collection of texts from 6 domains and 13 generators. The findings from CoAT underscored the difficulty of generalizing detection models to unseen generators and domains. The GenAI Content Detection Task at COLING 2025 [14] extended evaluation to a multilingual setting covering 21 languages, including Russian, with most top-performing systems relying on fine-tuned neural models to tackle domain shift [4]. These benchmarks show that domain adaptation and cross-lingual generalization remain open challenges; our work directly addresses the domain adaptation problem for the detection of Russian machine-generated texts, building on the foundations established by RuATD and CoAT while offering a computationally efficient ensemble solution.

2.3 N-gram Language Modeling

The foundation of our approach lies in n-gram language modeling. Chen and Goodman [2] provided the foundational empirical study of smoothing techniques for language modeling, establishing modified Kneser-Ney smoothing as the state-of-the-art method for n-gram models. Then Heafield [9] developed KenLM, an efficient implementation enabling fast and memory-efficient training and querying of such models. KenLM’s computational efficiency makes it particularly suitable for our ensemble approach, where multiple domain-specific models must be queried for each test text. The ability to build and query models quickly allows us also to scale our method to additional domains with minimal computational overhead. To the best of our knowledge, our work is the first to systematically apply domain-specific perplexity calibration using lightweight n-gram models specifically for AI-generated text detection, providing an interpretable and computationally efficient alternative to neural approaches.

3 Method

3.1 Overview

We propose an ensemble-based method for AI-generated text detection that operates without knowledge of the test text’s domain. The method consists of three main components: (1) training domain-specific language models, (2) calibrating classification thresholds for each domain, and (3) aggregating predictions through ensemble strategies.

Figure 1 provides a visual overview of our approach. We train separate trigram language models on human-written texts from multiple domains, calibrate thresholds for each model, and then use ensemble strategies to combine predictions for unseen texts.

3.2 Domain-Specific Language Models

Let $\mathcal{D} = \{d_1, d_2, \dots, d_M\}$ be a set of M domains. For each domain d , we train a trigram language model $\mathcal{M}_d$ using KenLM with modified Kneser-Ney smoothing. The training data consists exclusively of human-written texts from that domain. The model estimates conditional probabilities $P(t_i | t_{i-1}, t_{i-2})$ for each token t_i given its two predecessors, minimizing the negative log-likelihood of the training corpus.

For a text fragment F , perplexity under model $\mathcal{M}_d$ is defined as:

$$\text{PPL}_{\mathcal{M}_d}(F) = 2^{-\frac{1}{|F|} \sum_i \log_2 P_{\mathcal{M}_d}(t_i | t_{i-1}, t_{i-2})},$$

where $|F|$ is the text length in tokens. Lower perplexity indicates that the text is more predictable under that domain’s language model.

Rationale for n-gram models. We choose n-gram models over neural language models for three reasons: (1) they are computationally efficient and interpretable, (2) they capture domain-specific lexical and syntactic patterns without

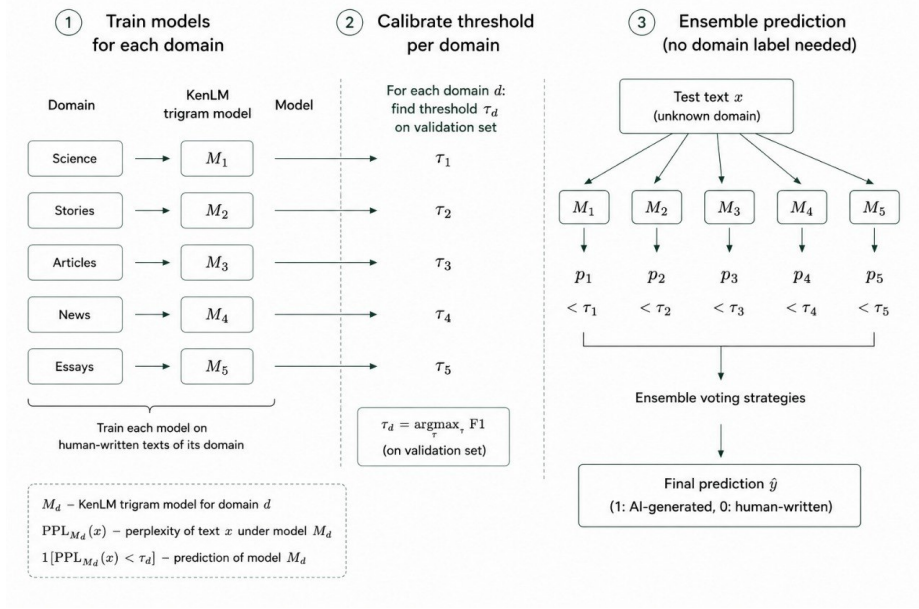


Fig. 1: Overview of the proposed method. For each domain, a KenLM model is trained on human-written texts. Perplexities are computed for the test text, and predictions from individual models are combined using ensemble strategies.

requiring extensive computational resources, and (3) they align with the observation that generative language models tend to sample from high-probability continuations [5], making n-gram statistics a natural baseline for detection.

3.3 Threshold Calibration

For each domain d , we calibrate a classification threshold τ_d using a validation set $\mathbf{V}_d$ containing balanced human-written and machine-generated texts from that domain. We search for the threshold that maximizes the F1-score:

$$\tau_d = \underset{\tau}{\operatorname{argmax}} \text{F1}(\mathbf{1}[\text{PPL}_{\mathcal{M}_d}(x) < \tau], y(x)),$$

where $y(x) = 1$ for machine-generated texts and $y(x) = 0$ for human-written texts, and F1 is the harmonic mean of precision and recall. In practice, we perform a grid search over 200 evenly spaced thresholds between the minimum and maximum perplexity values observed on the validation set.

The substantial variation in optimal thresholds across domains confirms that a universal threshold is inappropriate and that domain-specific calibration is essential.

3.4 Ensemble Strategies

For a test text T with unknown domain, we compute perplexities on trained models $p_d = \text{PPL}_{\mathcal{M}_d}(T)$ for all $d \in \mathcal{D}$. We then apply one of three ensemble strategies to combine individual predictions $h_d(T) = \mathbf{1}[p_d < \tau_d]$:

Strategy A: Simple Majority Voting. Each model votes based on its individual prediction. The final decision is determined by majority vote. This strategy is straightforward and assumes that most models will correctly classify the text.

Strategy B: Native Domain Detection. This strategy leverages the observation that a text from a particular domain yields minimal perplexity when evaluated on the model trained on that domain. We first identify the “native” domain $d^* = \arg \min_d p_d$, then use only the prediction of the corresponding model. This approach effectively performs domain detection before classification, using the most confident model for the given text.

Strategy C: Weighted Voting. Each model is assigned a weight proportional to its discriminative power, measured by the difference between its true positive rate and false positive rate on the validation set: $w_d \propto (\text{TPR}_d - \text{FPR}_d)$. Models that better distinguish between human and machine texts receive higher weights. The final prediction is a weighted combination, classifying the text as machine-generated if the total weight of machine votes exceeds 0.5.

Our method offers interpretability through clear per-domain predictions based on perplexity values, robustness via weighted voting that reduces the impact of weaker models, and scalability since adding a new domain requires only training one additional KenLM model and computing its validation statistics without retraining existing models. However, as raw perplexity values from different language models are not directly comparable in absolute values, we use two complementary strategies to be able to compare them: each model uses its own threshold calibrated independently on its validation set, and for native domain detection, we only use perplexity values to rank models and select the one with the minimum value, which is a relative comparison within the same text.

4 Experiments

4.1 Dataset Composition

We conduct experiments on a collection of human-written and machine-generated texts in Russian across several domains: science; fanfiction stories¹, articles [13], news from Gazeta² and Lenta.ru, and school graduation essays. Scientific documents were taken from AINL shared task dataset [1] and from Cyberleninka, a Russian scientific electronic library. The latter scientific subset, Lenta.ru news and essays were taken from **Ru-hard-detection dataset**³. Machine-generated

¹ IlyaGusev/ficbook

² IlyaGusev/gazeta

³ <https://github.com/CoffeBank/Ru-hard-detection-dataset>

texts were produced by the following LLMs: `llama-3.3-70b`, `qwen-2.5-72b`, `deepseek-v3`, `gpt-3.5`, `yandex-gpt-5-lite`, and `gemma-2-27b`. The same set of generators appears in the training, validation, and test splits, we did not include any held-out generators. The dataset is balanced with approximately 3,500 human-written and 3,500 machine-generated texts per domain. After filtering texts shorter than 50 words, the average length is 278.7 words for human and 265.1 for machine texts.

4.2 Experimental Setup

We treated subsets of scientific and news domain as a separate domain, thus for each experiment we trained seven KenLM models with trigram order ($n = 3$) and modified Kneser-Ney smoothing. The vocabulary is built from the training data, and we apply standard preprocessing, including lowercasing and tokenization.

4.3 Experiment 1: Single-Domain Training and In-Domain Testing

For each domain, we train a KenLM model exclusively on human-written texts from that domain and evaluate on a held-out test set from the same domain. Thresholds are optimized by F1-score on a validation set. Table 1 presents the optimal thresholds and corresponding F1-scores for each domain. The thresholds range from 8.28 (AINL science) to 12.10 (Lenta.ru news), demonstrating substantial domain-specific variation.

Table 1: Optimal thresholds and F1-scores for each domain (in-domain evaluation).

Domain	Optimal threshold τ_d	F1-score
Science (AINL)	8.28	0.85
Science (Cyberleninka)	11.56	0.83
Stories	12.09	0.78
Articles	9.60	0.73
News (Gazeta)	8.61	0.73
News (Lenta.ru)	12.10	0.75
Essays	8.91	0.72

Figure 2 provides a detailed visualization of the classification performance. The perplexity distributions show that machine-generated texts consistently exhibit lower perplexity than human-written texts, with the optimal threshold effectively separating the two classes. The ROC curves confirm strong discriminative power, with validation and test performance remaining consistent, indicating that the calibrated threshold generalizes well. These qualitative observations corroborate the quantitative results reported in Table 1.

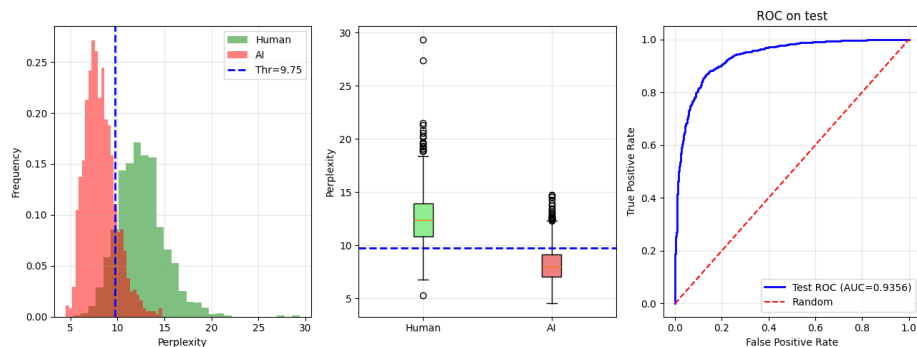


Fig. 2: Perplexity distributions for in-domain evaluation of science texts. The results demonstrate clear separation between human and machine texts and consistent performance across validation and test sets.

4.4 Experiment 2: Cross-Domain Testing

We evaluate each model on test sets from all other domains, using thresholds calibrated on the source domain. The results confirm that cross-domain performance drops substantially. For instance, the model trained on news achieves an F1-score of only 0.69 when tested on scientific texts, compared to 0.85 for in-domain evaluation. This provides strong evidence that perplexity thresholds are domain-dependent and cannot be directly transferred across domains.

4.5 Experiment 3: Unified Model

To establish a baseline for comparison, we combine all domains into a single balanced dataset and train a single KenLM model on the combined training set. This unified model achieves an accuracy of 0.68 and F1-score of 0.68, outperforming cross-domain transfer but still lagging behind in-domain performance. This indicates that domain heterogeneity remains a significant challenge even with a unified model.

4.6 Experiment 4: Tokenization and Pruning

We experiment with SentencePiece tokenization (vocabulary size of 8000) and n-gram pruning (`-prune 2 2 2`) to improve model stability across domains. Combined optimization yields a modest 1% improvement in accuracy (from 0.68 to 0.69), suggesting that while helpful, tokenization and pruning alone are insufficient to fully resolve domain shift.

4.7 Experiment 5: Ensemble Methods

We implement three ensemble strategies combining all domain-specific models. Table 2 presents the results.

Table 2: Comparison of ensemble methods on the mixed-domain test set.

Method	Accuracy	F1	Human / AI recall
Simple majority voting	0.76	0.75	0.79 / 0.72
Native domain detection	0.76	0.75	0.82 / 0.69
Weighted voting	0.76	0.75	0.79 / 0.72

All three ensemble strategies achieve comparable overall accuracy and F1-score, substantially outperforming the unified model (0.68). We note however that accuracy and F1 are the same for all three models, and the trade-off lies in class-specific performance: native domain detection favors human recall at the cost of AI recall with 0.82 and 0.69 recall, while simple majority voting and weighted provides more balanced performance, with recall 0.79 and 0.72, respectively.

5 Discussion

Our experiments confirm that perplexity-based detection suffers from significant domain shift. The variation in optimal thresholds across domains (8.28 to 12.10) is substantial, demonstrating that a single universal threshold is impractical. This finding agrees with the intuition that language models capture domain-specific patterns, and the predictability of a text is relative to the domain it belongs to.

The three ensemble strategies address this problem in complementary ways. Simple majority voting achieves balanced performance through democratic aggregation. Most models correctly classify any given text. Native domain detection illustrates that the model with minimal perplexity is likely trained on the text’s true domain, reaching high human recall. Weighted voting provides flexibility by emphasizing models with stronger discriminative power.

All ensembles outperform the unified model (0.76 vs. 0.68 accuracy), confirming that combining multiple domain-specific models is more effective than a single model trained on mixed data. This is particularly noteworthy because the ensemble method requires no knowledge of the test text’s domain, making it practical for real-world deployment.

Limitations. Our work is limited to Russian-language texts across five domains. Results may not directly generalize to other languages, particularly those with different morphological or syntactic characteristics. Additionally, we rely on n-gram models, which capture local lexical patterns but may miss long-range dependencies that neural models can exploit.

Future work. We plan to extend our approach to multiple languages and investigate adaptive threshold selection for unseen domains. Exploring the integration of neural language models as domain-specific baselines could potentially improve detection accuracy while maintaining the calibration framework.

6 Conclusion

In this work we addressed the issue of domain shift in perplexity-based AI-generated text detection. We have shown that perplexity values can vary substantially across domains and even within one domain on different subsets. To overcome this problem, we proposed an interpretable and computationally efficient ensemble approach with three aggregation strategies, combining domain-based lightweight KenLM models that achieve robust detection without requiring knowledge of text’s domain on the inference. Different way of aggregating model votes can either help to maximize human recall or ensure balanced performance, providing a functionality to adapt the tool for specific use-case. Our approach is flexible and scalable, allowing users to set up new domains without retraining other models.

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